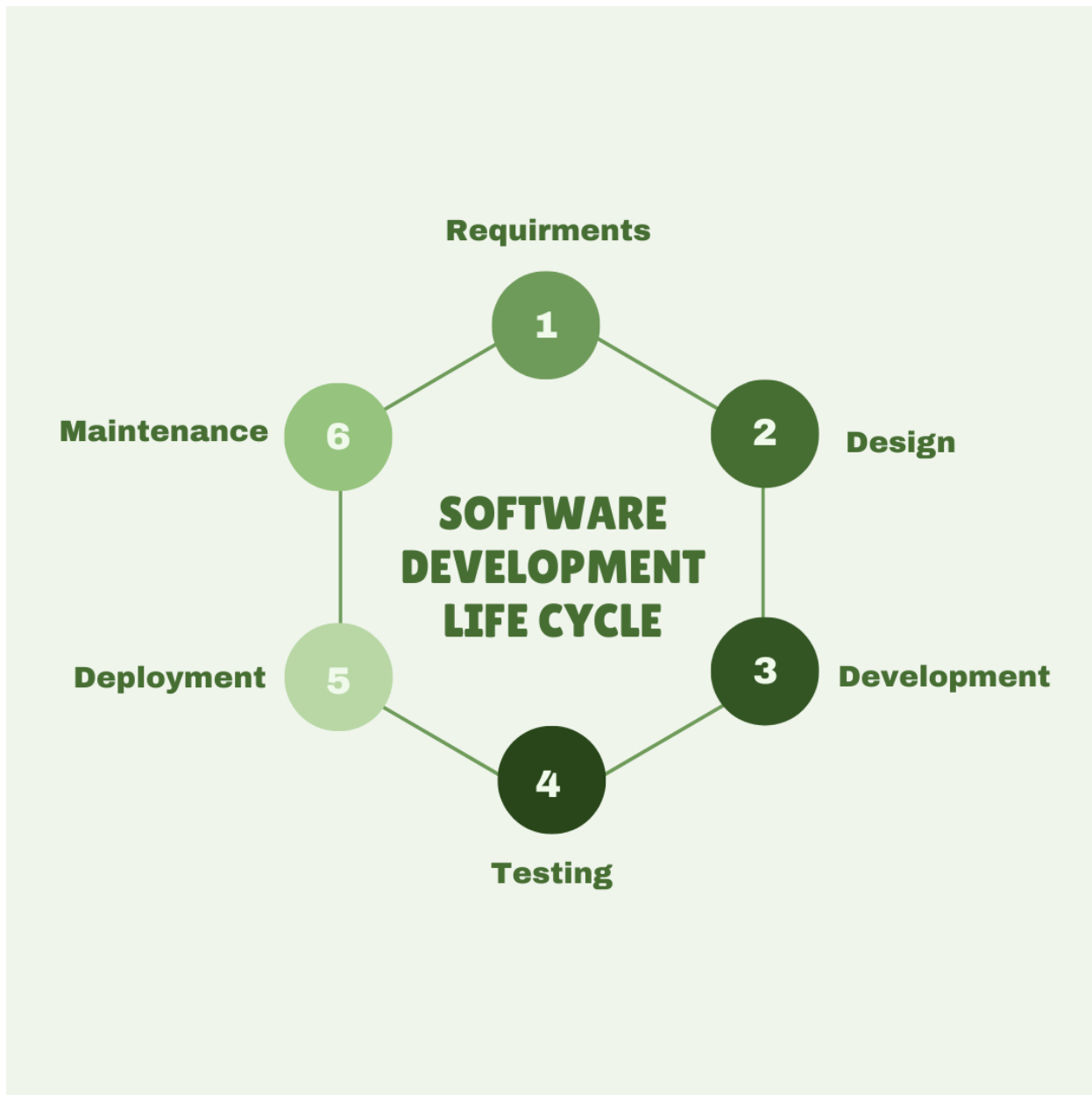


Software Development Life Cycle (SDLC) in the Age of AI

Assignment – 2

Q1. Draw the SDLC lifecycle diagram and briefly explain the workflow.

Answer :



Requirements: Understand the customer's needs and prepare requirements such as a BRD and an SRD.

Design: Decide how the system will be built, including architecture, database, API and UI/UX design.

Implementation: Developers write the actual source code. Code review and unit testing are also performed.

Testing: QA tests the software, reports defects and performs regression/retesting.

Deployment: After successful testing and approval, the software is released to the production environment.

Maintenance: After release, bugs are fixed, performance is improved and new features/security updates are added.

Practical AI Exercise

I am working on an Agile software development project for an E-commerce application.

Generate 5 realistic user stories using the format:
"As a [user], I want [feature], so that [benefit]."

Keep the user stories simple, clear, and suitable for a Sprint.

Here are 5 simple and realistic user stories for an E-commerce application, suitable for a Sprint:

1. As a customer, I want to search for products, so that I can quickly find the products I need.
2. As a customer, I want to add products to my shopping cart, so that I can purchase multiple products together.
3. As a customer, I want to view product details, so that I can decide whether a product is suitable for me.
4. As a customer, I want to place an order online, so that I can receive my selected products at my address.
5. As a customer, I want to track my order, so that I can know the current delivery status of my purchase.

ChatGPT was asked to generate five user stories for an E-commerce application

Take this user story:

"As a customer, I want to search for products so that I can easily find the products I need."

Create at least 5 clear and testable acceptance criteria for this user story.

Make sure the criteria include both positive and negative scenarios.

Here are 6 clear and testable acceptance criteria, including both positive and negative scenarios:

Acceptance Criteria — Product Search

#	Acceptance Criteria	Scenario
1	The customer can enter a product name in the search bar and click the Search button to see relevant products.	✓ Positive
2	The system displays products that match the customer's search keyword.	✓ Positive
3	The customer can search using partial product names or keywords and receive relevant results.	✓ Positive
4	If no product matches the search keyword, the system displays a clear message such as " No products found. "	✗ Negative
5	If the customer submits an empty search field, the system should not perform a search and should display a message such as " Please enter a search term. "	✗ Negative
6	If the customer enters unsupported or special characters as a search term, the system should handle the input safely and display appropriate results or a " No products found " message without crashing.	✗ Negative

ChatGPT was asked to convert one user story into at least five acceptance criteria

 Upgrade Create a Sprint Backlog for an E-commerce application based on the user stories above. Include: 1. Task 2. Description 3. Priority 4. Expected Outcome Make the backlog realistic for a development and QA team.																																																											
Below is a realistic Sprint Backlog for the E-commerce application, covering both Development and QA activities based on the user stories and acceptance criteria above. Sprint Backlog — E-commerce Application <table> <tr> <th>#</th><th>Task</th><th>Description</th><th>Priority</th><th>Expected Outcome</th></tr> <tr> <td>1</td><td>Design Search UI</td><td>Create the search bar, search button, and results section.</td><td>High</td><td>User can easily access and use the product search feature.</td></tr> <tr> <td>2</td><td>Develop Search Functionality</td><td>Implement product search using product names and keywords.</td><td>High</td><td>Relevant products are displayed when a customer searches.</td></tr> <tr> <td>4</td><td>Handle No Search Results</td><td>Display "No products found" when there are no matching products.</td><td>High</td><td>User receives a clear message when no products match.</td></tr> <tr> <td>5</td><td>Handle Empty Search</td><td>Validate the search field when the customer submits it without entering anything.</td><td>High</td><td>System displays "Please enter a search term" and does not perform an unnecessary search.</td></tr> <tr> <td>6</td><td>Handle Invalid Input</td><td>Validate special or unsupported characters and prevent application errors.</td><td>Medium</td><td>Invalid input is handled safely without crashing the application.</td></tr> <tr> <td>7</td><td>Create Search Test Cases</td><td>Prepare positive and negative test cases based on the acceptance criteria.</td><td>High</td><td>Search functionality has complete test coverage.</td></tr> <tr> <td>8</td><td>Execute Functional Testing</td><td>Test normal, partial, empty, and no-result searches.</td><td>High</td><td>Search feature works correctly for expected scenarios.</td></tr> <tr> <td>8</td><td>Execute Functional Testing</td><td>Test normal, partial, empty, and no-result searches.</td><td>High</td><td>Search feature works correctly for expected scenarios.</td></tr> <tr> <td>9</td><td>Execute Negative Testing</td><td>Test invalid inputs, special characters, and unexpected search values.</td><td>High</td><td>System handles invalid inputs safely and correctly.</td></tr> <tr> <td>10</td><td>Bug Reporting & Retesting</td><td>Report identified defects, verify fixes, and perform regression testing.</td><td>High</td><td>Confirmed defects are fixed without breaking existing functionality.</td></tr> </table>					#	Task	Description	Priority	Expected Outcome	1	Design Search UI	Create the search bar, search button, and results section.	High	User can easily access and use the product search feature.	2	Develop Search Functionality	Implement product search using product names and keywords.	High	Relevant products are displayed when a customer searches.	4	Handle No Search Results	Display " No products found " when there are no matching products.	High	User receives a clear message when no products match.	5	Handle Empty Search	Validate the search field when the customer submits it without entering anything.	High	System displays " Please enter a search term " and does not perform an unnecessary search.	6	Handle Invalid Input	Validate special or unsupported characters and prevent application errors.	Medium	Invalid input is handled safely without crashing the application.	7	Create Search Test Cases	Prepare positive and negative test cases based on the acceptance criteria.	High	Search functionality has complete test coverage.	8	Execute Functional Testing	Test normal, partial, empty, and no-result searches.	High	Search feature works correctly for expected scenarios.	8	Execute Functional Testing	Test normal, partial, empty, and no-result searches.	High	Search feature works correctly for expected scenarios.	9	Execute Negative Testing	Test invalid inputs, special characters, and unexpected search values.	High	System handles invalid inputs safely and correctly.	10	Bug Reporting & Retesting	Report identified defects, verify fixes, and perform regression testing.	High	Confirmed defects are fixed without breaking existing functionality.
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ChatGPT was asked to create a Sprint Backlog based on the E-commerce requirements.

Review the AI-generated output and improve it manually.

Explain

Q1. What the AI did well

Answer: The AI generated five relevant user stories and organised the Sprint Backlog into clear tasks. It also provided useful acceptance criteria for the product search feature.

Q2.What is the correction did you make?

Answer: I reviewed the AI-generated Sprint Backlog and found that the testing task was too general. I manually added specific testing tasks for testing empty search, testing special characters, testing search API errors, and testing invalid search

Manually Reviewed and Improved Sprint Backlog

Task	Description	Priority	Expected Outcome
Test Invalid Search	Test with non-existing product name	Medium	"No product found" message appears
Test Empty Search	Test search without entering text	Medium	Proper validation message appears
Test Special Characters	Test symbols and inputs	Medium	System handles input safely
Test search API Errors	Test server/API failure scenarios	High	Proper error message is shown

Q3. Why human review is still necessary?

Answer:

- AI may miss important requirements or cases.
- AI-generated information can sometimes be incorrect or incomplete.
- A human QA Engineer can verify that the output meets the actual business requirements.
- Human review helps identify missing test scenarios and potential risks.
- QA Engineers can manually improve and validate the AI-generated output.
- A human QA Engineer should make the final quality decision.

AI makes QA work faster and more efficient, while human judgment ensures accuracy and quality.